
Trial-Level Cross-Validation Can Invert Within- Versus Cross-Subject Comparisons

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Abstract

1 A common design in neural decoding compares a model trained on a subject's
2 own data against one trained on other subjects. The within-subject arm is usually
3 estimated by cross-validating over trials. We show that when a recording session
4 contains structure that happens to correlate with the label, this estimate is inflated
5 – and, because the cross-subject arm shares no session with its test data and cannot
6 be inflated the same way, the bias falls entirely on one side of the comparison. In
7 simulation with known ground truth the bias reverses the sign of the conclusion:
8 at a moderate nuisance level a trial-level split reports the within-subject decoder
9 beating cross-subject by +0.026 AUC when it in fact loses by 0.008. A label-
10 *independent* session offset produces no such bias, so the mechanism is specifically
11 session-level correlation with the label, not session identity. Nonlinear models ex-
12 ploit it $1.7\times$ more than linear ones, while ℓ_2 strength has no effect. On a widefield
13 calcium cohort (25 mice, DANDI:001425) the same substitution removes 30–73%
14 of a published-style asymmetry and eliminates one of four effects entirely, turning
15 +0.014 AUC at $p = 0.001$ into +0.003 at $p = 0.37$. On a second, independent
16 widefield cohort we find no such inflation, so the bias is a conditional hazard rather
17 than a universal one. We give the condition, the diagnostic, and the fix.

18 1 The problem

19 Many decoding studies ask whether a model trained on other individuals can substitute for one
20 trained on the subject, and answer it by comparing two accuracies: a within-subject estimate and a
21 cross-subject estimate. The comparison looks symmetric. It is not.

22 Trials recorded in one session are not independent. They share illumination and hemodynamic state,
23 slow arousal, electrode or lens drift, and simple temporal autocorrelation. A within-subject decoder
24 cross-validated over trials can therefore be trained on part of a session and tested on the rest of the
25 same session. A cross-subject decoder is trained on other individuals entirely and never shares a
26 session with its test data. Whatever advantage the shared session confers accrues to one arm only,
27 and therefore lands directly on the difference the study reports.

28 This note establishes three things: the precise structure that causes the bias, how large it gets, and
29 when it does *not* occur.

30 2 The mechanism, with ground truth

31 We simulate 12 subjects $\times$ 5 sessions $\times$ 100 trials in 44 dimensions:

$$x = \delta y w + \gamma y z_{s,k} + u_s + v_{s,k} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I). \quad (1)$$

Table 1: Simulation with known ground truth ($\sigma = 0.5$, $T = 100$, ℓ_2 logistic). *Inflation* is within-trial minus within-session. The last two columns are the within-minus-cross gap a study would report under each protocol; they differ in sign from $\gamma = 1$ onward.

γ	within (trial)	within (session)	cross	inflation	reported gap: trial / session
0.00	0.596	0.610	0.621	-0.014	-0.025 / -0.011
0.25	0.601	0.613	0.620	-0.012	-0.019 / -0.007
0.50	0.611	0.613	0.619	-0.003	-0.008 / -0.005
1.00	0.642	0.608	0.616	+0.034	+0.026 / -0.008
2.00	0.723	0.589	0.609	+0.135	+0.115 / -0.020

Here w is a decodable direction shared by all subjects, u_s a subject offset, $v_{s,k}$ a per-session offset of scale σ , and $z_{s,k}$ a direction drawn fresh for each session that correlates with the label *within that session only*. Labels are balanced Bernoulli, so $v_{s,k}$ carries no label information by construction and any accuracy attributable to it is leakage. We compare three estimators: within-subject with a trial-level 5-fold, within-subject with leave-one-session-out, and cross-subject (train on the other 11 subjects).

Session identity alone leaks nothing. Sweeping σ from 0 to 2 with $\gamma = 0$ produces inflation between -0.014 and +0.012 AUC, i.e. nothing systematic. This is worth stating because it is the intuitive story and it is wrong: a constant offset shifts a session’s cluster without moving the optimal boundary, so a linear decoder gains nothing from having seen the session before.

Session-level label correlation does. Holding $\sigma = 0.5$ and sweeping γ (Table 1), inflation is absent at $\gamma \leq 0.5$ and then grows sharply. The consequence is not merely that a number is too large. At $\gamma = 1$ the trial-level split reports the within-subject decoder ahead of cross-subject by +0.026 AUC; leave-one-session-out shows it is behind by 0.008. The bias reverses the sign of the reported comparison. At $\gamma = 2$ it manufactures +0.115 from a true -0.020.

Model class matters; regularization does not. At $\gamma = 1$, inflation is +0.0338, +0.0337 and +0.0338 for ℓ_2 logistic regression at $C = 0.01, 0.5$ and 100 – identical to four decimals across four orders of magnitude of penalty. A one-hidden-layer network on the same data inflates by +0.0564, $1.7\times$ more. The quantity that matters is the flexibility of the model class, not how hard it is penalized within a class. A capacity comparison run on a trial-level split therefore confounds capacity to fit the signal with capacity to exploit the leak.

3 A real instance

BraiDyn-BC [2] (DANDI:001425) provides 25 mice $\times$ ~ 15 daily sessions of cortex-wide widefield $\Delta F/F$ in 44 Allen-atlas parcels. We decode four task events and measure the personal-minus-pooled aging asymmetry – exactly the within- versus cross-subject shape described above – under both protocols on identical features.

Substituting leave-one-session-out for the trial-level split lowers the within-subject arm by +0.013, +0.010, +0.009 and +0.000 AUC across the four events, and leaves the cross-subject arm untouched, as predicted. The effect on the reported comparison (Table 2) is substantial: 30–73% of the asymmetry disappears, and for lever pull the entire effect does, turning +0.014 at $p = 0.001$ into +0.003 at $p = 0.37$. The remaining events survive at reduced magnitude.

The capacity prediction of Section 2 also holds here. Holding out sessions removes 43–73% of the CNN’s asymmetry against 30–48% of the linear model’s on the same events, so the higher-capacity model was the larger beneficiary of the leak – and a naive reading of the trial-split numbers would have concluded that the effect *grows* with capacity, which it does not.

4 A cohort where it does not occur

We ran the same substitution on an independent widefield cohort [1]: a different laboratory, a different parcellation (23 CCFv3 regions), spontaneous rather than cued behavior, and locomotion and

Table 2: BraiDyn-BC, $n = 25$ mice. Same features, same events, same statistic; only the split differs.

Event	trial-level split	session-held-out
Lever-pull	+0.014 ($p = 0.001$)	+0.003 ($p = 0.37$)
Lick	+0.028 ($p = 0.0002$)	+0.020 ($p = 0.002$)
Reward	+0.032 ($p = 0.0001$)	+0.029 ($p = 0.0005$)
Tone	+0.022 ($p = 0.005$)	+0.015 ($p = 0.08$)
Pooled	+0.024	+0.017

pupil-derived arousal as targets. Each animal is one continuous recording, so the analogue of a session-held-out split is a contiguous-block split.

We found no inflation. For locomotion the within-subject arm changed by -0.005 AUC and for arousal by -0.046 ; in both cases the shuffled estimate was *lower* than the blocked one, and for arousal this held in 6 of 6 animals. The within-minus-cross gap was accordingly not overstated in this cohort.

We report this because it bounds the claim. The bias is not an automatic consequence of splitting trials; it requires the session to contain label-correlated structure for the decoder to absorb. Section 2 makes that condition explicit, and a cohort can fail to meet it.

5 Recommendation

Hold out whole recording sessions when estimating the within-subject arm of a within- versus cross-subject comparison. The cost is a slightly noisier estimate; the benefit is that the two arms are exposed to the same information.

Because the hazard is conditional, we also suggest reporting both numbers when the comparison is load-bearing. The difference between them is itself the diagnostic: it is an estimate of how much session-level label-correlated structure the data contain, costs one extra fit to obtain, and would have flagged the effect in Section 3 immediately.

Limitations. We have one positive cohort and one negative one, so we cannot say how common the hazard is in practice, only that it is real, that it can invert a conclusion, and that it is not universal. Our account of why the second cohort is unaffected is post hoc: we predicted inflation there and did not find it, and while the mechanism in Section 2 is consistent with the null, we have not independently measured label-correlated session structure in either cohort and shown it differs. A direct measurement of γ in real data, and a survey of published within-versus-cross designs, are the natural next steps.

References

- [1] H. Benisty et al. Rapid fluctuations in functional connectivity of cortical networks encode spontaneous behavior. *Nature Neuroscience*, 27, 2024. figshare 175317.
- [2] A. Kondo et al. BraiDyn-BC: a cortex-wide widefield calcium imaging dataset during a cued lever-pull operant task. *Scientific Data*, 12, 2025. DANDI:001425.